

Practicum CFD 2026-2027

Lecturer: R. Luppès (r.luppès@rug.nl)

Practicum 1

deadline: Monday, September 14th, 17:00

- Working in pairs (groups of max. 2 students) is allowed.
- Each practicum is closed with an oral discussion supported by a (short) written report of the findings. Make the report by pen/paper (readable) or with L^AT_EX, Word, etc.
- Always use Python or Matlab/Octave for the programs that you develop.
- Save your programs in separate files, such that it is easy to demonstrate them.
- The use of AI tools such as ChatGPT is not allowed!
- Be sure that you have discussed the exercises and programs before the deadline!

Exercise 1 – Compose program

Compose a program that solves the steady convection-diffusion equation

$$U \frac{d\phi}{dx} = k \frac{d^2\phi}{dx^2} \quad \text{on} \quad 0 \leq x \leq L, \quad \text{with} \quad \phi(0) = \phi_0, \quad \phi(L) = \phi_1.$$

Take into account the following:

- As input variables, the program should use values of U , k , ϕ_0 , ϕ_1 and the number of segments N covering the domain $[0, L]$.
- Various discretisation methods should be possible, e.g. central, (smart-)upwind.
- Very fine grids should be possible, e.g. $h = L/N$, with $N = 100,000$.

Hence, you cannot store the full 2D coefficient matrix A and you cannot use linear solver commands (in Matlab) such as `inv(A)*b` or `A\b`. Instead, you should only store the 3 diagonals of A and use the TDMA algorithm as provided (in Matlab), or use `scipy.sparse.diags` (in Python).

Remark: during other practica we will include more discretisation options (e.g. 2nd order upwind, Quick), non-equidistantly spaced grids and the unsteady term $d\phi/dt$.

Exercise 2 – Wiggles

Consider the case $L=10$, $U=50$, $k=10^{-3}$, with boundary conditions $\phi_0=0$, $\phi_1=20$.

- When does central discretisation result in a smooth solution without wiggles? Compare your "experimental" result with the theoretical grid-size limit.
- Does upwind discretisation always result in a smooth solution?
Find a relation between the obtained boundary layer thickness and the grid size, both experimentally and theoretically, and compare these.
When is the solution "accurate enough"? How would you define "enough"?
- The boundary condition $\phi(10)=20$ forces the solution towards a fixed value at $x=10$. The alternative boundary condition (so called Neumann condition) $\phi'(10)=0$ gives the system more flexibility. This might enhance the occurrence of wiggles, although the solution reduces to $\phi(x)=0$ (constant).
Include this boundary condition in your program of Exercise 1, and check whether there are more/larger wiggles at a certain grid size, and whether the grid-size limit for central discretisation changes.
- Now take $\phi'(10)=20$ as boundary condition. This also gives the system more freedom, but will not give a trivial solution. Again check whether this affects the occurrence of wiggles.
- Explain the situation when $\phi'(0) = \phi'(10) = 0$. What happens to the matrix in this case?

P.T.O.

Exercise 3 – Accuracy study

Consider the original problem with Dirichlet conditions $\phi(0)=\phi_0$ and $\phi(L)=\phi_1$. Take "nice" values for L , U , k and ϕ_0 , ϕ_1 , such that $Pe = 500$.

- (a) Show by means of a number of numerical simulations (i.e. runs of your program of Exercise 1) that the central method has 2nd order accuracy everywhere, and the (regular) upwind method has 1st order accuracy everywhere. Use the analytic solution as reference solution.
- (b) How can you determine the order of accuracy numerically when the analytic solution is not known? Try this.

Exercise 4 – Artificial diffusion

Files and commands Required files: `cfid_1.m`, `cfid_upw.m`, `ns_exact.m`
The program is started in Matlab with the command `cfid_1`.

Description

Consider the inhomogeneous convection-diffusion equation ($U = 1$)

$$\frac{d\phi}{dx} - k \frac{d^2\phi}{dx^2} = S \quad \text{on } 0 \leq x \leq 1, \quad \text{with } \phi(0) = 0, \quad \phi(1) = 1.$$

The right-hand side is given by

$$S(x) = \begin{cases} 2 - 5x & , \quad 0 \leq x \leq 0.4, \\ 0 & , \quad 0.4 < x \leq 1. \end{cases}$$

In this exercise an upwind discretisation of the above equation is applied. The discrete solution for three values of the diffusion coefficient k is plotted in one picture. Also the exact solution for one value of k can be included in this picture.

Questions

- (a) Firstly, study the discrete solution for a grid with $N = 20$ segments, at the following values of k : $k = 0.1$, 0.01 and 0.001 . These values correspond with mesh-Péclet numbers $P = 0.5$, 5 and 50 , respectively. One would expect (or at least hope) that these solutions tend to the (analytical) solution for $k = 0$. Hence, compare the discrete solutions to the exact solution at $k = 0$ (ignore the Matlab warnings concerning division by zero). What do you observe; is there any such relation?
- (b) Next, compare the same three discrete solutions at $k = 0.1$, 0.01 and 0.001 (again for $N = 20$) to the exact solution at $k = 0.025$. Now, there seems to be some relation between the upwind limit for $k \rightarrow 0$ and the exact solution at $k = 0.025$. Explain this relation.
- (c) Finally, study once more the upwind solutions at $k = 0.1$, 0.01 and 0.001 , but now on a grid with $N = 40$ segments. Which exact solution, i.e. at which value of k , corresponds with the upwind limit for $k \rightarrow 0$?